TVA

Tennessee Valley Authority, Sequoyah Nuclear Plant, P.O. Box 2000, Soddy Daisy, Tennessee 37384

October 7, 2016

10 CFR 50.73

ATTN: Document Control Desk
U.S. Nuclear Regulatory Commission
Washington, D.C. 20555-0001

Sequoyah Nuclear Plant, Units 1 and 2
Renewed Facility Operating License Nos. DPR-77 and DPR-79
NRC Docket Nos. 50-327 and 50-328

Subject: **Licensee Event Report 50-327 and 50-328/2016-008-00, Closed Fire Damper Renders Both Trains of the Control Room Emergency Ventilation System Inoperable**

The enclosed licensee event report provides details concerning discovery of a closed fire protection damper associated with the Control Room Emergency Ventilation System (CREVS) that resulted in both trains of CREVS being declared inoperable. This report is being submitted in accordance with 10 CFR 50.73(a)(2)(i)(B), as any operation or condition which was prohibited by the plant's Technical Specifications and in accordance with 10 CFR 50.73(a)(2)(v)(D), as an event or condition that could have prevented the fulfillment of a safety function of structures or systems that are needed to mitigate the consequences of an accident.

There are no regulatory commitments contained in this letter. Should you have any questions concerning this submittal, please contact Michael McBrearty, Site Licensing Manager, at (423) 843-7170.

Respectfully,

Preston P. Pratt
Plant Manager
Sequoyah Nuclear Plant

Enclosure: Licensee Event Report 50-327 and 50-328/2016-008-00
cc: NRC Regional Administrator – Region II
NRC Senior Resident Inspector – Sequoyah Nuclear Plant

NRC FORM 366 (11-2015) U.S. NUCLEAR REGULATORY COMMISSION	APPROVED BY OMB: NO. 3150-0104 EXPIRES: 10/31/2018
LICENSEE EVENT REPORT (LER)	Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Privacy and Information Collections Branch (T-5 F53), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by internet e-mail to Infocollects.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOB-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

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4. TITLE
Closed Fire Damper Renders Both Trains of the Control Room Emergency Ventilation System Inoperable

5. EVENT DATE			6. LER NUMBER			7. REPORT DATE			8. OTHER FACILITIES INVOLVED	
MONTH	DAY	YEAR	YEAR	SEQUENTIAL NUMBER	REV NO.	MONTH	DAY	YEAR	FACILITY NAME: Sequoyah Nuclear Plant Unit 2	DOCKET NUMBER: 05000328
08	10	2016	2016 -	008 -	00	10	07	2016	FACILITY NAME: NA	DOCKET NUMBER

9. OPERATING MODE	11. THIS REPORT IS SUBMITTED PURSUANT TO THE REQUIREMENTS OF 10 CFR §: *(Check all that apply)*			
1	☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)
	☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)
	☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)
	☐ 20.2203(a)(2)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)
10. POWER LEVEL	☐ 20.2203(a)(2)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(v)(A)	☐ 73.71(a)(4)
	☐ 20.2203(a)(2)(iii)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(v)(B)	☐ 73.71(a)(5)
	☐ 20.2203(a)(2)(iv)	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(1)
100	☐ 20.2203(a)(2)(v)	☐ 50.73(a)(2)(i)(A)	☒ **50.73(a)(2)(v)(D)**	☐ 73.77(a)(2)(i)
	☐ 20.2203(a)(2)(vi)	☒ **50.73(a)(2)(i)(B)**	☐ 50.73(a)(2)(vii)	☐ 73.77(a)(2)(ii)
		☐ 50.73(a)(2)(i)(C)	☐ OTHER Specify in Abstract below or in NRC Form 366A	

12. LICENSEE CONTACT FOR THIS LER

LICENSEE CONTACT	TELEPHONE NUMBER *(Include Area Code)*
Scott T Bowman	423-843-6910

13. COMPLETE ONE LINE FOR EACH COMPONENT FAILURE DESCRIBED IN THIS REPORT

CAUSE	SYSTEM	COMPONENT	MANU-FACTURER	REPORTABLE TO EPIX	CAUSE	SYSTEM	COMPONENT	MANU-FACTURER	REPORTABLE TO EPIX

14. SUPPLEMENTAL REPORT EXPECTED	15. EXPECTED SUBMISSION DATE	MONTH	DAY	YEAR
☐ YES *(If yes, complete 15. EXPECTED SUBMISSION DATE)* ☒ **NO**				

ABSTRACT *(Limit to 1400 spaces, i.e., approximately 15 single-spaced typewritten lines)*

On August 10, 2016, at 1000 Eastern Daylight Time (EDT), the Maintenance Instrument Group (MIG) performed a surveillance instruction (SI) that resulted in a fire protection damper associated with the Control Room Emergency Ventilation System (CREVS) actuating to a closed position. On August 11, at 1015 EDT, the fire damper was discovered in the closed position. The closure of the fire protection damper resulted in both trains of CREVS being inoperable due to the loss of air circulation and habitability of the Technical Support Center and Relay Room. Both units entered Technical Specification Limiting Condition for Operation (LCO) 3.7.10, Conditions A and G. Condition G requires immediate entry into LCO 3.0.3. At 1159 EDT, on August 11, actions were taken to open the fire damper and both trains of CREVS were declared operable. This allowed both units to exit LCO 3.0.3 and LCO 3.7.10, Conditions A and G.

The cause of the event was determined to be inadequate breaker alignment instructions in the SI. Corrective actions include revising the SI to require the associated breaker to be open prior to performance of the SI.

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NARRATIVE

I. PLANT OPERATING CONDITIONS BEFORE THE EVENT

At the time of the event, Sequoyah Nuclear Plant (SQN) Unit 1 and Unit 2 were in Mode 1 at 100 percent rated thermal power.

II. DESCRIPTION OF EVENTS

A. Event:

On August 10, 2016, at 1000 Eastern Daylight Time (EDT),the Maintenance Instrument Group (MIG) began performing a surveillance instruction (SI) associated with a functional test of the fire detection system [EIIS: IC] and peripheral devices. During performance of the SI, MIG received the expected response from the test, which was an alarm [EIIS: ALM]. Multiple times during the test, MIG was instructed by the SI to acknowledge the alarm (by pressing the “send” key) and continue with testing. At some point during performance of the SI, a fire protection damper [EIIS: DMP] associated with the Control Room Emergency Ventilation System (CREVS) [EIIS: VI] received an actuation signal and actuated to a closed position. While closing, the damper actuator broke its associated fusible links, as designed. This was not an anticipated result of the SI, and therefore, went undiscovered.

On August 11, 2016, at 1015 EDT, while performing a walkdown of the Relay Room, the fire protection damper was discovered in the closed position. The closure resulted in both trains of CREVS being inoperable due to the loss of air circulation and habitability of the Technical Support Center (TSC) and Relay Room. Both units entered Technical Specification (TS) Limiting Condition for Operation (LCO) 3.7.10, Conditions A and G. Condition G requires immediate entry into LCO 3.0.3. At 1159 EDT, on August 11, actions were taken to block open the fire damper and both trains of CREVS were declared operable. This allowed both units to exit LCO 3.0.3 and LCO 3.7.10, Conditions A and G.

An 8-hour non-emergency event notification (EN 52172) was made to the NRC in accordance with 10 CFR 50.72(b)(3)(xiii), as any event that results in a major loss of emergency assessment capability, offsite response capability, or offsite communications capability, and 10 CFR 50.72(b)(3)(v)(D), as an event or condition that could have prevented fulfillment of a safety function of structures or systems that are needed to mitigate the consequences of an accident. This LER documents the reportable event under 10 CFR 50.73(a)(2)(v)(D).

Additionally, it was determined that both trains of CREVS were inoperable from August 10, 2016, at 1000 to August 11, 2016, at 1159. LCO 3.7.10, Condition G, Required Action G.1 requires immediate entry into LCO 3.0.3. LCO 3.0.3 requires the unit to be in MODE 3 within 7 hours and MODE 4 within 13 hours. Because both trains of CREVS were inoperable for approximately 26 hours and remained in MODE 1, this is a condition prohibited by TS and is therefore being

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reported in accordance with 10 CFR 50.73(a)(2)(i)(B), as any operation or condition which was prohibited by the plant's TS.

B. Status of structures, components, or systems that were inoperable at the start of the event and contributed to the event:

No inoperable structures, components, or systems contributed to this event.

C. Dates and approximate times of occurrences:

Date/Time (EDT)	Description
08/10/16, 1000	MIG began performance of an SI associated with a functional test of the fire detection system and peripheral devices. During performance of the SI, a fire protection damper associated with the CREVS received an actuation signal and actuated to a closed position.
08/10/16, 1400	MIG completed the SI with all acceptance criteria satisfied.
08/11/16, 1015	The fire protection damper was discovered in the closed position. The closure resulted in both trains of CREVS being inoperable due to the loss of air circulation and habitability of the Technical Support Center and Relay Room. Both units entered LCO 3.7.10, Conditions A and G and LCO 3.0.3.
08/11/16, 1159	The fire protection damper was blocked open and secured in its required position to support the design safety function. This restored both CREVS trains to operable status. Both units exited LCO 3.7.10, Conditions A and G and LCO 3.0.3.

D. Manufacturer and model number of each component that failed during the event:

There was no component that failed during the event. While closing, the damper actuator broke its associated fusible links, as designed.

E. Other systems or secondary functions affected:

There were no other systems or secondary functions affected by this event.

F. Method of discovery of each component or system failure or procedural error:

While performing a walkdown of the Relay Room, the fire protection damper was discovered in the closed position.

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G. The failure mode, mechanism, and effect of each failed component, if known:

There was no component that failed during the event. While closing, the damper actuator broke its associated fusible links, as designed.

H. Operator actions:

Both trains of CREVS were declared inoperable due to the loss of air circulation and habitability of the TSC and Relay Room. Both units entered LCO 3.7.10, Conditions A and G and LCO 3.0.3.

I. Automatically and manually initiated safety system responses:

There were no automatic or manual system responses associated with this event.

III. CAUSE OF THE EVENT

A. The cause of each component or system failure or personnel error, if known:

There was no component or system failure associated with this event.

The cause of the event was determined to be inadequate breaker alignment instructions in the SI. Had the associated breaker been open during the SI test, the fire protection damper would not have received an actuation signal and would not have closed.

B. The cause(s) and circumstances for each human performance related root cause:

There was no identified human performance related root cause.

IV. ANALYSIS OF THE EVENT

The CREVS provides a protected environment from which occupants can control the unit following an uncontrolled release of radioactivity, hazardous chemicals, or smoke. The CREVS consists of two independent, redundant trains that recirculate and filter the air in the control room envelope (CRE) and a CRE boundary that limits the inleakage of unfiltered air. Each CREVS train consists of a high efficiency particulate air (HEPA) filter, an activated charcoal adsorber section for removal of gaseous activity (principally iodines), and a fan. Ductwork, valves or dampers, doors, barriers, and instrumentation also form part of the system.

The CREVS is an emergency system, parts of which may also operate during normal unit operations in the standby mode of operation. Actuation of the CREVS places the system in the emergency mode of operation. Actuation of the system to the emergency radiation state of the emergency mode of operation, closes the unfiltered outside air intake and unfiltered exhaust dampers, and aligns the system for recirculation of the air within the CRE through the redundant trains of HEPA filters and the charcoal adsorbers. The emergency radiation state also initiates pressurization and filtered ventilation of the air supply to the CRE.

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The closed fire protection damper resulted in the CREVS being unable to provide air circulation to the TSC and Relay Room (both rooms are located within the CRE). This could have prevented the TSC from being habitable had plant conditions required activation of the TSC. The event would not have challenged the habitability of the main control room (MCR).

V. ASSESSMENT OF SAFETY CONSEQUENCES

There were no actual safety consequences as a result of this event. Engineering judgment determined the event could have resulted in a slight increase in dose to the MCR operators; however, the dose would have remained below 10 CFR 100 limits.

A. Availability of systems or components that could have performed the same function as the components and systems that failed during the event:

There were no components or systems that failed during the event.

B. For events that occurred when the reactor was shut down, availability of systems or components needed to shutdown the reactor and maintain safe shutdown conditions, remove residual heat, control the release of radioactive material, or mitigate the consequences of an accident:

The event did not occur when the reactor was shutdown.

C. For failure that rendered a train of a safety system inoperable, an estimate of the elapsed time from discovery of the failure until the train was returned to service:

The elapsed time from discovery of both trains of CREVS being inoperable until both trains were restored to operable status was approximately 104 minutes.

VI. CORRECTIVE ACTIONS

This event was entered into the Tennessee Valley Authority Corrective Action Program under Condition Reports (CR) 1201905.

A. Immediate Corrective Actions:

Actions were taken to block the fire protection damper in the open position. Additionally to address the extent of condition, actions were taken to inspect other fire protection dampers.

B. Corrective actions to reduce probability of similar events occurring in the future:

Corrective actions include revising the SI to require the associated breaker to be open prior to performance of the SI.

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VII. ADDITIONAL INFORMATION

A. Previous similar events at the same plant:

A review of SQN LERs identified an event that resulted in both trains of CREVS being declared inoperable. LER 327/2016-003 reported a condition that could have prevented the fulfillment of a safety function of structures or systems that are needed to mitigate the consequences of an accident when both trains of CREVS were declared inoperable because of an inoperable Control Room Envelope. The cause of the event was the bottom screw in the door latch guard appears to have stripped loose and fallen into the rubber seal around the base of the MCR door frame preventing closure of the door. Additionally, LERs 327/2013-004, 327/2015-001, and 328/2015-001 report events associated with procedures.

B. Additional Information:

None.

C. Safety System Functional Failure Consideration:

This condition resulted in a safety system functional failure.

D. Scrams with Complications Consideration:

There was no scram associated with this event.

VIII. COMMITMENTS

None.